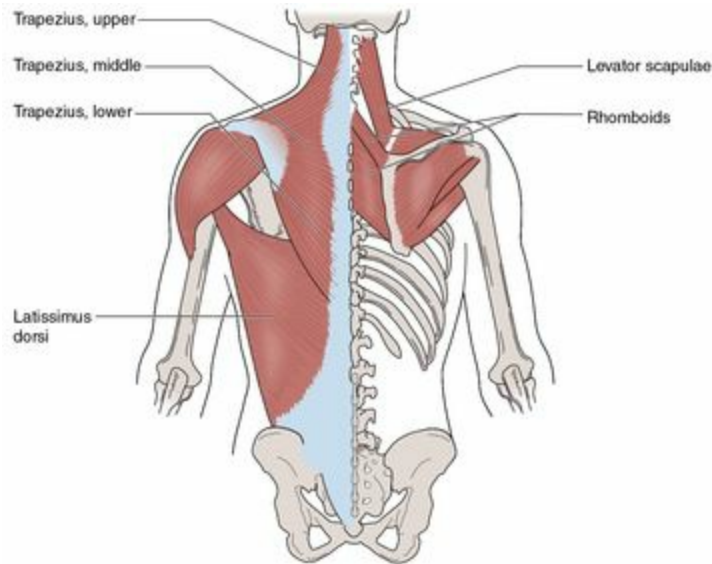


Figure 3.36 The latissimus dorsi originates from each lumbar spinous process via the lumbodorsal fascia and inserts on the humerus to perform both lumbar extension and stabilization roles.



Exercise for the Extensor Muscles of the Low Back

Clinical Relevance

Research has shown that the thoracic extensors (longissimus thoracis pars thoracis and iliocostalis lumborum pars thoracis) that attach in the thoracic region are actually the most efficient lumbar extensors because they have the largest moment arms as they course over the lumbar region. For this reason, it is time to revisit the clinical practice of isolating muscle groups—in this case, the lumbar extensors for the lumbar spine. Specifically, what are referred to as the lumbar extensors (located in the lumbar region) contribute only a portion of the total lumbar extensor moment. Training the lumbar extensor mechanism must involve the extensors that attach to the thoracic vertebrae, whose bulk of contractile fibers lies in the thoracic region but whose tendons pass over the lumbar region and have the greatest mechanical advantage of all the lumbar muscles. Thus, exercises to isolate the lumbar muscles cannot be justified anatomically or from a motor control perspective because all “players in the orchestra” must be challenged during training.

Another important clinical issue involves the anatomical features of the

extensors. Although the lumbar sections of the longissimus and iliocostalis muscles that attach to the lumbar vertebrae create extensor torque, they also produce large posterior shear forces to support the shearing loads that develop during torso flexion postures. Some therapists unknowingly disable these shear force protectors by having patients fully flex their spines during exercises, creating myoelectric quiescence in these muscles, or by recommending the pelvic tilt during flexing activities such as lifting. A discussion of this functional anatomy is critical for developing the strategies for injury prevention and rehabilitation described later in this book.

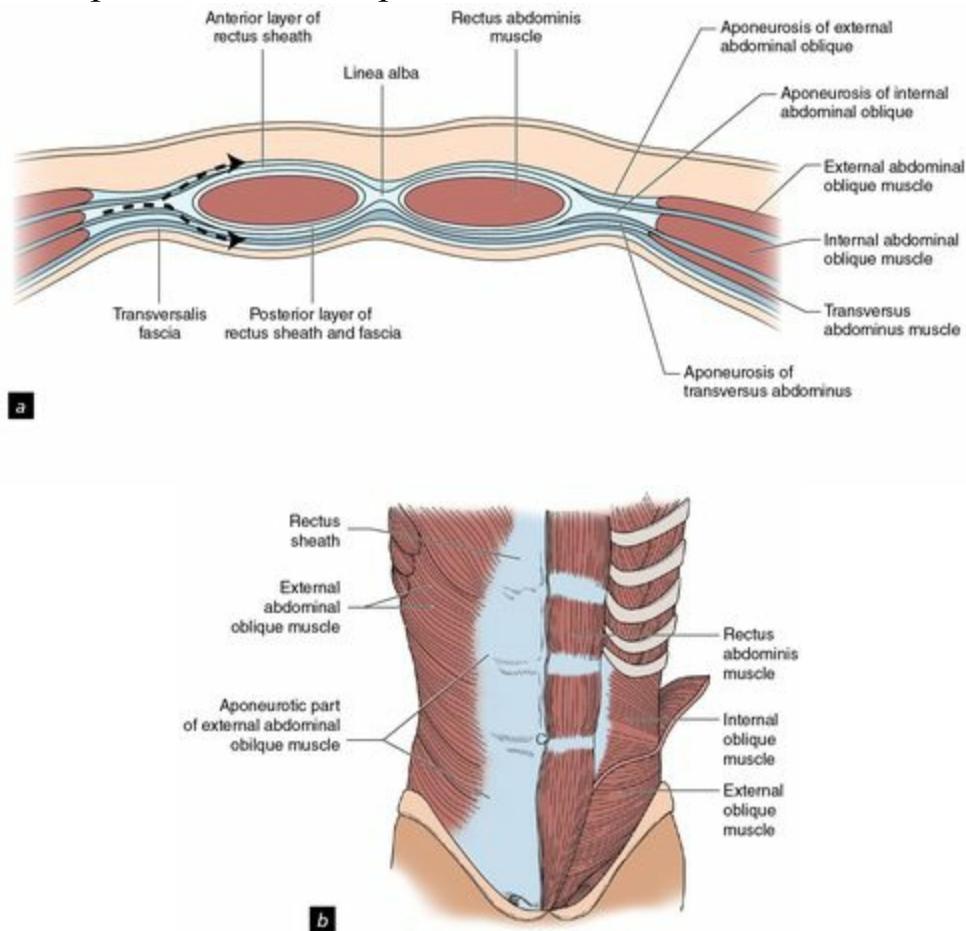
Anterior Muscles

This section addresses several important aspects of lumbar mechanics in which the abdominal muscles are involved.

Abdominal Fascia

The abdominal fascia contains the rectus abdominis and connects laterally to the aponeurosis of the three layers of the abdominal wall. Its functional significance is made more important by connections of the aponeurosis with pectoralis major, together with fascial elements that cross the midline to transmit force to the fascia (and abdominal muscles) on the opposite side of the abdomen (Porterfield and DeRosa, 1998) (see [figure 3.37](#), a and b). Such anatomical features underpin and justify exercises (detailed later) that integrate movement patterns that simultaneously challenge the abdominal muscles, the spine, and the shoulder musculature.

Figure 3.37 The abdominal fascia connects the obliques of the abdominal wall with (a) rectus abdominis and, to a lesser extent, (b) pectoralis major and helps to transmit hoop stresses around the abdomen.



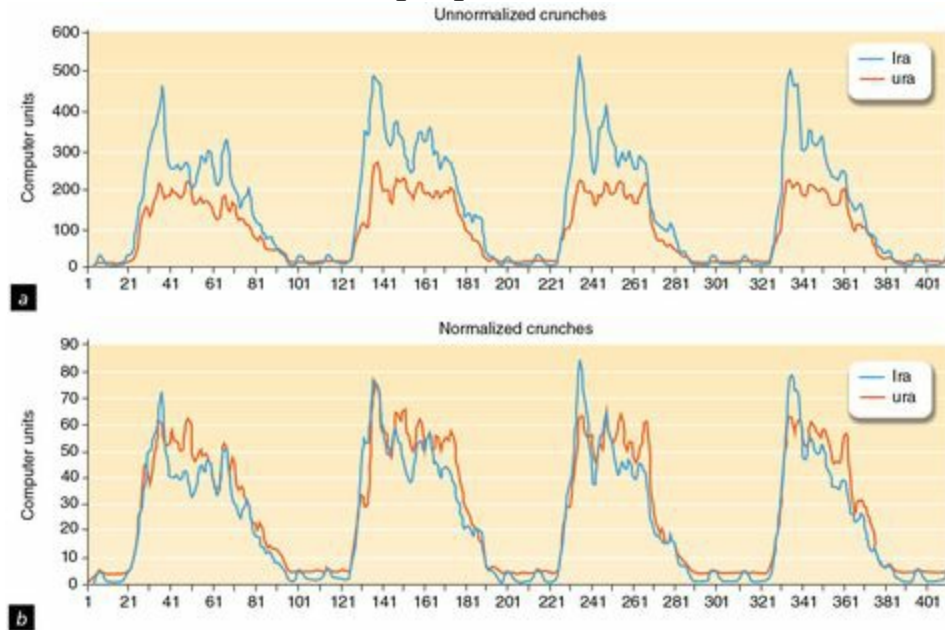
Although many classic anatomy texts consider the abdominal wall to be an important flexor of the trunk, the rectus abdominis appears to be the major trunk flexor—and the most active during sit-ups and curl-ups (Juker, McGill, and Kropf, 1998). Muscle activation amplitudes obtained from both intramuscular and surface electrodes over a variety of tasks are shown in [table 4.4](#) in chapter 4. It is interesting to consider why the rectus abdominis is partitioned into sections rather than being a single long muscle, given that the sections share a common nerve supply and that a single long muscle would have the advantage of broadening the force–length relationship over a greater range of length change. Perhaps a single muscle would bulk upon shortening, compressing the viscera, or be stiff and resistant to bending. Not only does the sectioned rectus abdominis limit bulking upon shortening, but the sections

also have a bead effect, which allows bending at each tendon to facilitate torso flexion–extension or abdominal distension or contraction as the visceral contents change volume.

The beaded rectus also performs another role—the lateral transmission of forces from the oblique muscles that form a continuous hoop around the abdomen (Porterfield and DeRosa, 1998). The intermuscular tendons and fascia prevent the fibers of rectus from being ripped apart laterally from these hoop stresses. This aspect of abdominal mechanics is elucidated further in the next section (Abdominal Wall), which addresses the forces developed in the oblique muscles.

Another clinical issue is the controversy regarding upper and lower abdominal muscles. Although the obliques are regionally activated (and have functional separation between upper and lower regions), all sections of the rectus are activated together at similar levels during flexor torque generation. A significant functional separation does not appear to exist between the upper and lower rectus (Lehman and McGill, 2001) in most people. Research reporting differences in upper and lower rectus activation sometimes suffers from the absence of normalization of the EMG signal during processing. Briefly, researchers have used raw amplitudes of myoelectric activity (in millivolts) to conclude that there is more, or less, activity relative to other sections of the muscle, but the magnitudes are affected by local conductivity characteristics. Thus, amplitudes must be normalized to a standardized contraction and expressed as a percentage of this activity (rather than in millivolts) (see [figure 3.38](#)). Researchers may also have inadvertently monitored pyramidalis (an optional muscle at the base of the rectus), which would cloud interpretation.

Figure 3.38 (a) Studies that have reported an upper and lower rectus abdominis generally evaluated unnormalized EMG signals—for example, in the study of men performing crunches. (b) However, when the signals are normalized properly, the apparent difference disappears. Although there are regional activation zones in the abdominal obliques, there is little evidence to suggest that a functional upper and lower rectus abdominis exists in most people.



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Upper and Lower Rectus

Clinical Relevance

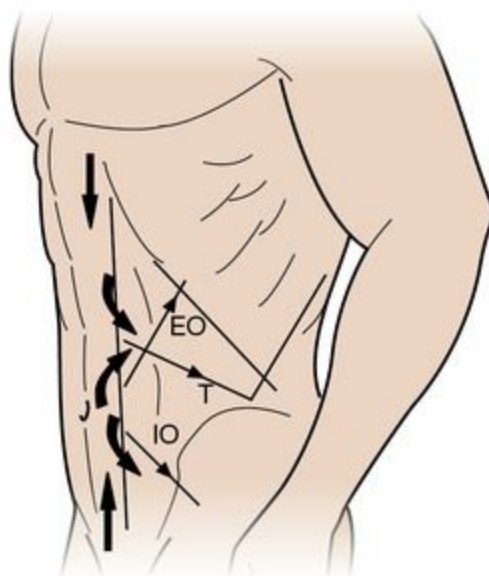
Although we have seen separation in neural drive between upper and lower sections of rectus abdominis in a few Middle Eastern–style belly dancers (out of a larger cohort), these abilities and moves are rare in everyday life. (We made these observations only during very low-level contractions in which the muscles were moved in the absence of substantial flexion torque or muscle load.) In fact, a distinct upper and lower rectus does not exist in most people. Once force is required, the rectus appears to act as a cable with active

tension along its entire length. Thus, training the rectus for nearly everyone can be accomplished with a single exercise. This is not true for the obliques, because they have several neural compartments—lateral, medial, upper, and lower.

Abdominal Wall

The three layers of the abdominal wall (external oblique, internal oblique, and transverse abdominis) perform several functions. All three are involved in flexion and appear to have their flexor potential enhanced because of their attachment to the linea semilunaris (see [figure 3.39](#)) (McGill, 1996), which redirects the oblique muscle forces down the rectus sheath to effectively increase the flexor moment arm. The obliques are involved in torso twisting (McGill, 1991a, 1991b) and lateral bend (McGill, 1992) and appear to play a role in lumbar stabilization because they increase their activity, to a small degree, when the spine is placed under pure axial compression (Jucker, McGill, and Kropf, 1998). (This functional notion is developed later in this chapter.) The obliques are also involved in challenged lung ventilation, assisting with what is called active expiration (Henke et al., 1988). The important functional implication of active expiration is explained in McGill (2014). For example, when athletes grunt at the beginning of the expiration phase, it enhances spring stability.

Figure 3.39 The oblique muscles (EO: anterior portion of external oblique; IO: anterior portion of internal oblique; T: transverse abdominis) transmit force along their fiber lengths and then redirect force along rectus abdominis via their attachment to the linea semilunaris to enhance their effective flexor moment arm.



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Researchers have focused a lot of attention on the transverse abdominis for two reasons. First, some believe it is involved in spine stability through its beltlike containment of the abdomen and also in the generation of intra-abdominal pressure (IAP). Richardson and colleagues (1999) nicely summarized the second reason for the focused attention on this muscle by suggesting that transverse has a delayed onset of activation time in some with back troubles, prior to a rapid arm movement. Their hypothesis is that the trunk must first be made stiff and stable. It appears that this is evidence that an aberrant motor pattern exists in rapid activity, but according to Silfies and colleagues (2009), all muscles exhibit these patterns in different subgroups of back pain. Further we cannot find evidence that this motor pattern happens bilaterally, nor could others (e.g., Allison, Morris, and Lay, 2008), implying that it has little to do with stability. The question remains as to whether this is important in more normally paced activities. Delayed onset confirms motor

deficits, but a 10 to 30 ms onset delay would be irrelevant during a normal movement in which these muscles are continually cocontracted to ensure stability. Contrary to what is commonly heard in exercise discussions, the evidence does not suggest that transverse abdominis is not recruited. It is also interesting that, again, contrary to popular thought, the transverse and internal oblique are very similar in their fiber orientation. In our very limited intramuscular EMG electrode work (Juker, McGill, and Kropf, 1998), we saw a high degree of coupling between these two muscles in a wide variety of nonballistic exertions (no doubt some myoelectric cross-talk existed). We have been unable to find people who can or cannot activate transverse beyond a level of 2% of maximum. Nonetheless, several groups (Cresswell, Oddsson, and Thorstensson, 1994, and Hodges and Richardson, 1996, are the most experienced) have noted the extra activity of transverse when IAP is elevated, but also that all of the abdominal muscles show this activity. The combination of transverse activation (and almost always concomitant oblique activity) with elevated IAP enhances stability, without question. In fact, Cholewicki, Juluru, and McGill (1999) concluded that building IAP on its own adds spine stability.

In the broader functional perspective, the components of the abdominal muscles (rectus, obliques, and transverse) work together but also independently. Although a variety of sources have provided myoelectric evidence from a variety of tasks, an anatomical and functional interpretation is needed. As noted earlier, the obliques differentially activate to create twisting torque and can enhance flexion torque. Rectus is primarily a flexor—in fact, those people who have a great deal of motor control in the abdominal muscles can preferentially activate each muscle (see [figure 3.40](#), a-d). Finally, some have suggested an upper and lower partitioning of the abdominal muscles. As previously mentioned, this impression is probably an artifact resulting from poor EMG technique; there does not appear to be a functional upper and lower rectus (Lehman and McGill, 2001; Vera-Garcia, Grenier, and McGill, 2000). On the other hand, regional differences do exist in the obliques; some sections can be preferentially recruited both medially and laterally together with upper and lower portions. Finally, the obliques, together with transverse abdominis, form a containing hoop around the entire abdomen; the anterior of the hoop is composed of the abdominal fascia, and the posterior is composed of the lumbodorsal fascia. The resulting hoop stresses and stiffness assist with spine stability.

Figure 3.40 Some trained people have the ability to differentially activate specific portions of the abdominal musculature. This sequence shows (a) an inactive abdominal wall, (b) the abdominal wall ballooned, and (c) the contraction of transverse abdominis, which draws and hollows the wall. (Note that other muscles must relax to do this, which compromises both stability and strength.) (d) Placing the hands on the thighs and pushing allows a flexor moment to develop in which good muscular control is able to activate just the rectus abdominis with the previously activated transverse and little oblique activity.

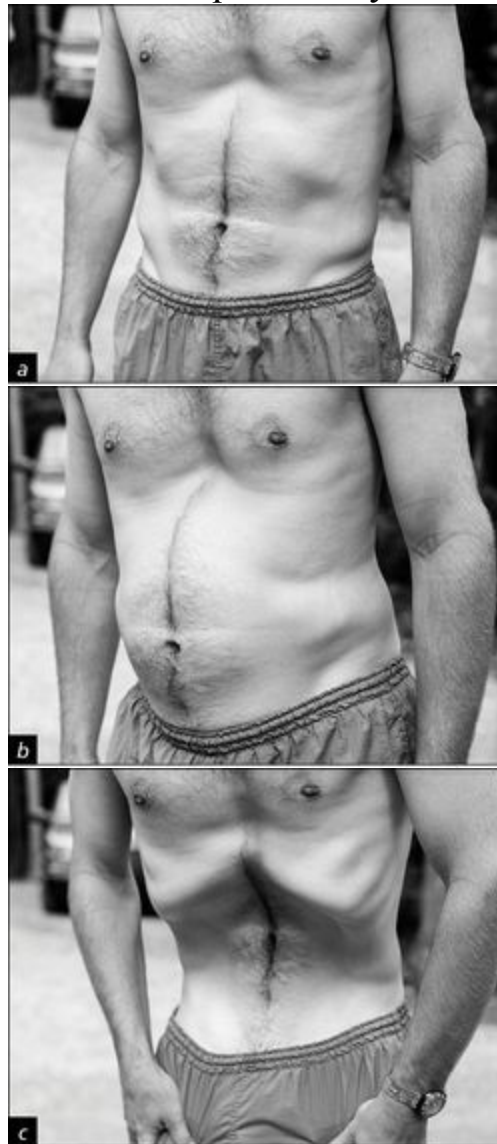




Photo from Stuart McGill

Abdominal Muscle Exercises

Clinical Relevance

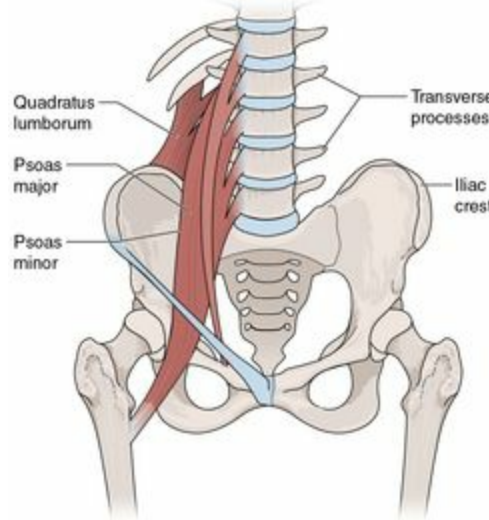
The functional divisions of the abdominal muscles justify the need for several exercise techniques to challenge them in all of their roles: moment generation, spine stability, and heavy breathing. Although the obliques are regionally activated, with several neural compartments, there appears to be no functional separation of upper and lower rectus abdominis. Thus, a curl-up exercise activates all portions of the rectus abdominis. However, upper and lower portions of the oblique abdominal muscles are activated separately depending on the demands placed on the torso. Finally, detailed examination of the fascial connections reveals force transmission among the shoulder musculature, the spine, and the abdominal muscles, justifying exercises incorporating larger movement patterns for the advanced patients who can bear the higher loads. Techniques to obtain high performance from the abdominal muscles enhanced with power breathing are beyond the scope of this book, but are documented in *Ultimate Back Fitness and Performance* (McGill, 2014).

Psoas

The psoas, a muscle that crosses the spine and hip, is unique for many reasons. Although it has been claimed to be a major stabilizer of the lumbar

spine, I believe that this claim needs interpretation. Although the psoas complex attaches to T12 and to every lumbar vertebra on its course over the pelvic ring (see [figure 3.41](#)), its activation profile (see Andersson et al., 1995; Juker, McGill, and Kropf, 1998; and Juker, McGill, Kropf, and Steffen, 1998, for indwelling EMG data) is not consistent with that of a spine stabilizer (in the purest sense); rather, it indicates that the role of the psoas is purely as a hip flexor. During our work to implant intramuscular electrodes in this muscle, we performed the insertion technique several times on ourselves. The first time the electrodes were in my own psoas, I tried several low back exertions in an attempt to activate it (flexion, extension, side-bending, and so on.). None of these really caused the psoas to fire. Simply raising the leg while standing, with hip flexion, caused massive activation, clearly indicating that the psoas is a hip flexor.

Figure 3.41 The psoas attaches to each lumbar vertebra (lateral vertebral body and transverse process) where each of these laminae of psoas fuse and form a common tendon that courses through the iliopectineal notch to the femur. The quadratus lumborum attaches each transverse process with the ribs and iliac crest, forming the guy wire support system.



After this pilot work, in a larger study we found that any task requiring hip flexion involved the psoas. Although we obtained this impression from studying the activation profile of the psoas, other considerations stem from its architecture. Why does the psoas traverse the entire lumbar spine and, in fact, course all the way to the lower thoracic spine? Why not let just the iliacus perform hip flexion? If only the iliacus were to flex the hip, the pelvis would be torqued into an anterior pelvic tilt, forcing the lumbar spine into extension. These forces are buttressed by the psoas, which adds stiffness between the pelvis and the lumbar spine. In effect, it can be thought of as a spine stabilizer but only in the presence of significant hip flexor torque. Also, an activated and stiffened psoas will contribute some shear stiffness to the lumbar motion segment—but once again, only when hip flexor torque is required. The fact is that the psoas and iliacus are two separate muscles (see Santaguida and McGill, 1995), functionally, architecturally, and neurally. There is no such thing as an iliopsoas muscle!

In addition, some clinical discussion has been centered on the issue of whether the psoas is an internal or external rotator of the femur and hip. Although it provides some anatomical advantage during external hip rotation, Juker, McGill, Kropf, and Steffen (1998) observed only small activation bias

during hip rotation tasks. However, this may have been due to the need for significant hip stabilization, resulting in substantial hip cocontraction. In a recent study, we examined a matched cohort of workers—half of whom had a history of disabling low back troubles, and half of whom had never missed work. Those who had a history, but were asymptomatic at the time of the test, had a significant loss of hip extension and hip internal rotation (but more external rotation). This is an interesting observation, given the extensive clinical discussions regarding the so-called tight psoas. Even though we do not fully understand the neuromechanics, this muscle as a clinical concern is worth studying further.

Psoas Function

Myoelectric evidence and anatomical analysis suggest that the psoas major acts primarily to create hip flexion torque, that its activation is minimally linked to spine demands, and that it imposes substantial lumbar spine compression when activated. Caution is advised when training this muscle because of the substantial compression penalty imposed on the spine when the psoas is activated.

Quadratus Lumborum

The quadratus lumborum (QL) is another special muscle for several reasons. First, the architecture of this muscle provides a stabilizing role by attaching to each lumbar vertebra, effectively buttressing adjacent vertebrae bilaterally, and by attaching to the pelvis and rib cage (see [figure 3.41](#)). Specifically, the fibers of the QL cross-link the vertebrae and have a large lateral moment arm via the transverse process attachments. Thus, by its design the QL could buttress shear instability and be effective in stabilizing all loading modes. Typically, under compressive load, the first mode of buckling instability is lateral (Lucas and Bresler, 1961); the QL can play a significant role in local lateral buttressing. Also, the QL hardly changes length during any spine motion (see [table 3.2](#) and McGill, 1991b), suggesting that it contracts virtually isometrically. Further insight into its special

function comes from an earlier observation that the motor control system involves this muscle together with the abdominal wall when stability is required in the absence of major moment demands.

Quadratus Lumborum and Spine Stabilization

Clinical Relevance

The QL appears to be highly involved with stabilization of the lumbar spine, together with other muscles, suggesting that a clinical focus on this muscle is warranted. Exercises emphasizing activation of the QL while sparing the spine are described in part III, Low Back Rehabilitation.

The QL muscle appears to be active during a variety of flexion-dominant, extensor-dominant, and lateral bending tasks. (Note that myoelectric access to the QL is quite tricky, and it is difficult to confirm where the intramuscular electrodes are within the muscle. Certainly, our techniques on this muscle were not very precise. In addition, they tend to migrate upon contraction, further clouding interpretation of the signal.) Andersson and colleagues (1996) found that the QL did not relax with the lumbar extensors during the flexion-relaxation phenomenon. The flexion-relaxation phenomenon is an interesting task because there are no substantial lateral or twisting torques and the extensor torque appears to be supported passively, further suggesting some stabilizing role for the QL.

In another experiment (note again that our laboratory techniques to obtain QL activation were rather imprecise at the time), subjects stood upright with a bucket in each hand. We incrementally increased the load in each bucket (resulting in progressively more spine compression). Our data suggest that the QL increased its activation level (together with the obliques) as more stability was required (McGill, Juker, and Kropf, 1996b). This task creates a special situation because only compressive loading is applied to the spine in the absence of any bending moments. In summary, the strength of the evidence from several perspectives leaves one to conclude that the QL performs a very special stabilizing role for the lumbar spine in a wide variety of tasks. Most recently, the Hodges group (Park et al., 2013) strengthened the concept that there can be a redistribution of muscle activity in pain. In particular, there is a tendency for activity to migrate extensor activity to the

quadratus lumborum. Given the number of patients with back pain who have perceived spasm and pain in this muscle, experimental follow-up is recommended.

Muscle Summary

This section provides an overview of the roles of the muscles of the trunk in supporting postures and moving and stabilizing the lumbar spine. The posterior muscles were presented in four large functional groups. The muscles in the deepest group (the small rotators) appear to act as position sensors rather than as torque generators. The more superficial extensors (multifidus and iliocostalis lumborum and longissimus thoracis) fall into three categories to do the following:

- Generate large extension moments over the entire lumbar region
- Generate posterior shear
- Affect and control only one or two lumbar segments

The roles of the abdominal muscles in trunk flexion and trunk stabilization were highlighted together with the roles of the psoas and QL. Clearly, many muscles play a large role in ensuring sufficient stiffness to bear load and create movement, while protecting the low back from injury. Chapter 10 applies these findings to exercise regimens for those with low back pain.

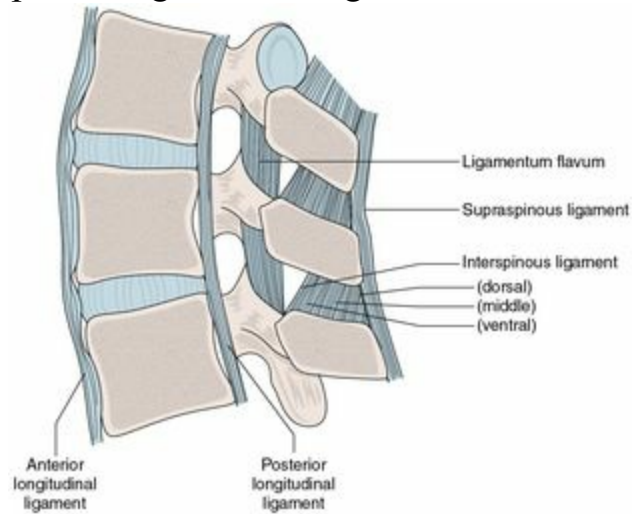
Ligaments

When the lumbar spine is neither flexed nor extended (neutral lordosis), only muscle contributions need be considered in the mechanics to support the spine. However, as the spine flexes, bends, and twists, passive tissues are stressed; the resultant forces of those tissues change the interpretation of injury exacerbation, the discussion of clinical issues, or both. For this reason, I introduce the mechanics of passive tissues in this section, followed by some examples illustrating their effects on clinical mechanics. Also fascinating is the distribution of mechanoreceptors documented in every lumbar ligament and fascia. Solomonow and colleagues' (2000) evidence suggests a significant proprioceptive role for spinal ligaments.

Longitudinal Ligaments

The vertebrae are joined to form the spinal column by two ribbon-like ligaments, the anterior longitudinal and the posterior longitudinal ligaments, which assist in restricting excessive flexion and extension (see [figure 3.42](#)). Both ligaments have bony attachments to the vertebral bodies and collagenous attachments to the annulus. Very little evidence exists for the presence of mechanoreceptors in these ligaments. Posterior to the spinal cord is the ligamentum flavum, which is characterized by a composition of approximately 80% elastin and 20% collagen, signifying a very special function for this ligament. It has been proposed that this highly elastic structure, which is under pretension throughout all levels of flexion, acts as a barrier to material that could buckle and encroach on the cord in some regions of the range of motion.

Figure 3.42 Major lumbar ligaments. Note the controversy surrounding the interspinous ligament in figures 3.43 and 3.44.



Interspinous and Supraspinous Ligaments

The interspinous and supraspinous ligaments are often classed as a single structure in anatomy texts, although functionally they appear to have quite different roles. The interspinous ligaments connect adjacent posterior spines but are not oriented parallel to the compressive axis of the spine. Rather, they have a large angle of obliquity, which has been a point of contention. For years most anatomy books have shown these ligaments with an oblique angle that would cause posterior shear of the superior vertebrae (see [figure 3.43](#)). This error is believed to have originated around the turn of the century. The hypothesis is that an artist held the vertebral section upside down when drawing; other artists simply copied the previous art rather than look at a spine. This was corrected by Heylings (see [figure 3.44](#)), noting that indeed these ligaments have the obliquity to resist posterior shear of the superior vertebrae—but also impose anterior shear forces during full flexion (Heylings, 1978). Although many anatomy textbooks suggest that this ligament serves to protect against excessive flexion (based on erroneous anatomy), I do not believe that this notion is correct. Heylings (1978) suggested that the ligament acts like a collateral ligament similar to those in the knee, whereby the ligament controls the vertebral rotation as it follows an arc throughout the flexion range. This in turn helps the facet joints remain in

contact, gliding with rotation. Furthermore, with its oblique line of action, the interspinous ligament protects against posterior shearing of the superior vertebrae and is implicated in an injury scenario discussed later in this chapter.

Figure 3.43 For most of the past 100 years, many anatomical artists have drawn the interspinous ligament upside down, as shown in this example. Such drawings have caused the ligament function to be misinterpreted as that of a supporter of anterior shear, which is incorrect.

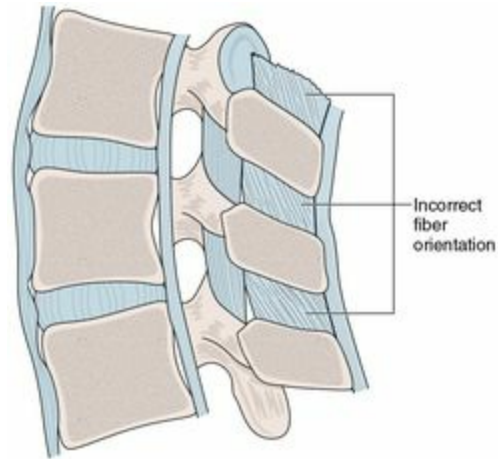
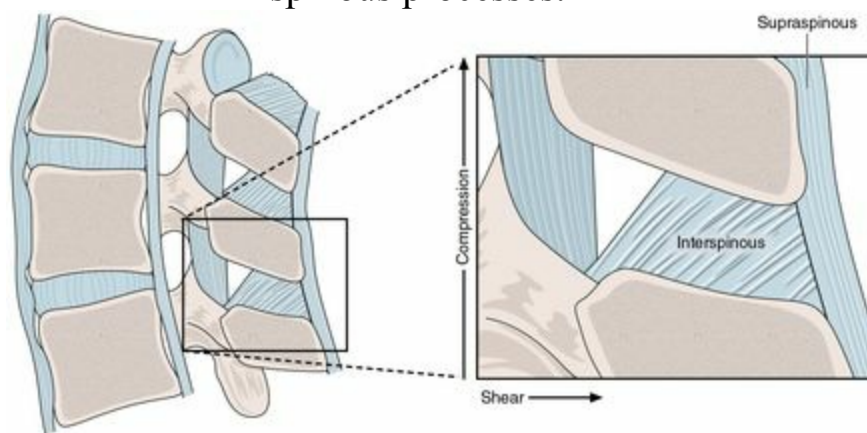


Figure 3.44 The interspinous ligament runs obliquely to the compressive axis and thus has limited capacity to check flexion rotation of the superior vertebrae. Rather, the interspinous ligament may act as a collateral ligament, controlling vertebral rotation and imposing anterior shear forces on the superior vertebrae. (ZG) Zygapophyseal joint or facet joint, (L2), (L3) lumbar spinous processes.



In contrast to the interspinous ligament, the supraspinous ligament is aligned more or less parallel to the compressive axis of the spine, connecting the tips of the posterior spines. It appears to provide resistance against excessive forward flexion. Finally, both supraspinous and interspinous ligaments have an extensive network of free nerve endings (type IV receptors) (Yahia, Newman, and Rivard, 1988) together with Ruffini corpuscles and Pacinian corpuscles (Jiang et al., 1995; Yahia, Newman, and Rivard, 1988). Yahia and colleagues (1988) and Solomonow and colleagues (2000) suggested a proprioceptive role for the ligaments to prevent excessive strain in fully flexed postures and—given their architecture—quite possibly when under excessive shear load.

Interspinous Ligament Architecture

Clinical Relevance

The interspinous ligament appears to provide a collateral action as it guides the sliding motion of the facet joints and checks posterior shear of the superior vertebra. The flip side of this functional role is that during full flexion the superior vertebra is sheared anteriorly, often adding to the reaction

shear forces produced in a forward bending posture. Therapeutic exercise recommending full spine flexion stretches must consider the resultant shearing forces imposed on the joint by interspinous ligament strain. Too often, even patients with shear pathology—for example, those with spondylolisthesis—are prescribed flexion stretches. This appears to be ill-advised.

Other Ligaments in the Thoracolumbar Spine

Other ligaments in the thoracolumbar spine include the intertransverse ligaments and the facet capsule.

- **Intertransverse ligaments.** These ligaments span the transverse processes and have been argued to be sheets of connective tissue rather than true ligaments (Bogduk and Twomey, 1991). In fact, Bogduk and Twomey suggested that the intertransverse ligament membrane forms a septum between the anterior and posterior musculature that is an embryological holdover from the development of these two sections of muscle.
- **Facet capsule.** The facet capsule consists of connective tissue with bands that restrict both the joint flexion and the distraction of the facet surfaces that result from axial twisting. The ligaments that form the capsule have been documented to be rich in proprioceptive organs—Pacinian and Ruffini corpuscles (Cavanaugh et al., 1996; McLain and Pickar, 1998)—and have been observed to respond to multidirectional stress (Jiang et al., 1995), at least in cats.

Normal Ligament Mechanics and Injury Mechanics

Determining the roles of ligaments has involved qualitative interpretation using their attachments and lines of action together with functional tests in which successive ligaments were cut and the joint motion was reassessed. Early studies attempting to determine the amount of relative contribution of each ligament to restricting flexion were performed on cadaveric preparations that were not preconditioned prior to testing. This usually entails a loading regimen so that the cadaveric specimens better reflect in vivo behavior.

Specifically, the investigators did not take into account the fact that upon death, discs, being hydrophilic, increase their water content and consequently their height. The swollen discs in cadaveric specimens produced an artificial preload on the ligaments closest to the disc, inappropriately suggesting that the capsular and longitudinal ligaments are more important in resisting flexion than they actually are in vivo. For this reason, these early data describing the functional roles of various ligaments were incorrect. The work of Sharma, Langrama, and Rodriguez (1995) showed that the major ligaments for resisting flexion are the supraspinous ligaments. Those early studies showing that the posterior longitudinal ligament and the capsular ligaments are important for resisting flexion did not employ the necessary preconditioning just discussed.

Mechanical failure of the ligaments is worth considering. King (1993) noted that soft tissue injuries are common during high-energy, traumatic events such as automobile collisions. Our own observations on pig and human specimens loaded at slow load rates in bending and shear modes suggest that excessive tension in the longitudinal ligaments causes avulsion or bony failure near the ligament attachment site. Noyes, De Lucas, and Torvik (1994) noted that slow strain rates (0.66%/s) produced more ligament avulsion injuries, whereas fast strain rates (66%/s) resulted in more midligamentous failure, at least in monkey knee ligaments. The clinical results of Rissanen (1960), however, showed that approximately 20% of randomly selected cadaveric spines possessed visibly ruptured lumbar interspinous ligaments, whereas the dorsal and ventral portions of the interspinous ligaments and the supraspinous ligaments were intact.

Patterns of Ligament Injury

Clinical Relevance

Torn spinal ligaments appear to be the result of ballistic loading—particularly slips and falls or traumatic sporting activity with the spine at its end range of motion. Those with recently developing spine symptoms and accompanying spine instability can often recount prior incidents in which the ligaments could have been damaged. If the traumatic event occurs at work but the delayed sequelae develop later during an event at home, is this compensable? Arguing these types of questions requires a solid

understanding of injury mechanisms.

These results could be considered to represent the living population. Given the oblique fiber direction of the interspinous complex (see [figure 3.44](#)), a very likely scenario of interspinous ligament damage is falling and landing on one's behind, driving the pelvis forward on impact and creating a posterior shearing of the lumbar joints when the spine is fully flexed. The interspinous ligament is a major load-bearing tissue in this example of high-energy loading characterized by anterior shear displacement combined with full flexion. Considering the available data, I believe that disruption of the ligaments of the spine, particularly the interspinous complex, is not likely during lifting or other normal occupational activities. Rather, ligament damage seems to occur primarily during traumatic events, as described earlier. The subsequent joint laxity is well known to accelerate arthritic changes (Kirkaldy-Willis and Burton, 1992). What has been said in reference to the knee joint, that ligament damage marks the beginning of the end, is also applicable to the spine in terms of being the initiator of a cascade of degenerative change.

Chronic loading of the posterior ligaments, at subfailure loads, occurs during repeated bending and flexing of the spine. Solomonow's group has documented in many studies the changes in proprioceptive function as the ligaments are repeatedly stretched. Their recent work has shown elevated proinflammatory cytokines 7 hours after a repeated loading session and suggested that acute inflammation conditions can occur that can become chronic inflammation with further exposure to ligament loading (D'Ambrosia et al., 2010).

Lumbodorsal Fascia (LDF)

Although a functional interpretation of the lumbodorsal fascia (LDF) (also called the thoracolumbar fascia by some) is provided later, a short anatomical description is given here. First, the fascia has bony attachments on the tips of the spinous process (except the shorter L5 in many people) and to the posterior superior iliac spines (PSIS). Some fascial connections cross the midline, suggesting some force transmission, thus completing the hoop around the abdomen with the previously described abdominal fascia anteriorly. The transverse abdominis and internal oblique muscles obtain their posterior attachment to the fascia, as does the latissimus dorsi over the upper regions of the fascia. The fascia, in wrapping around the back, forms a compartment around the lumbar extensors (the multifidus and pars lumborum groups of the iliocostalis and longissimus) and has been implicated in compartment syndrome (Carr et al., 1985; Styf, 1987) (see [figure 3.45](#)). Recent studies attribute various mechanical roles to the LDF. In fact, some have recommended lifting techniques based on these hypotheses. However, are they consistent with experimental evidence? Gracovetsky, Farfan, and Lamy (1981) originally suggested that lateral forces generated by the internal oblique and transverse abdominis muscles are transmitted to the LDF via their attachments to the lateral border, and that the fascia could support substantial extensor moments. They hypothesized that this lateral tension on the LDF increased longitudinal tension by virtue of the collagen fiber obliquity in the LDF, causing the posterior spinous processes to move together, resulting in lumbar extension. This proposed sequence of events formed an attractive proposition because the LDF has the largest moment arm of all the extensor tissues. As a result, any extensor forces within the LDF would impose the smallest compressive penalty to vertebral components of the spine.

Figure 3.45 Collagen fiber arrangement in the LDF binds the lumbar extensor muscles and tendons from the thoracic muscles together as they course to the sacral attachments. Thus, one of the functions of the LDF appears to be acting as an extensor muscle retinaculum—and a natural abdominal–back belt.



Photo from Stuart McGill

However, three studies, all published at about the same time, collectively challenged the viability of this hypothesis: Tesh and colleagues (1987) performed mechanical tests on cadaveric material; Macintosh, Bogduk, and Gracovetsky (1987) recognized the anatomical inconsistencies with the abdominal activation; and McGill and Norman (1988) tested the viability of LDF involvement with the latissimus dorsi as well as with the abdominal muscles (see [figure 3.46](#)). These collective works show that the LDF is not a significant active extensor of the spine. Nonetheless, the LDF is a strong tissue with a well-developed lattice of collagen fibers, suggesting that its function may be that of an extensor muscle retinaculum (Bogduk and Macintosh, 1984), or nature's back belt. In addition, the fascia does contain both Ruffini and Pacinian corpuscles together with diffuse innervation (Yahia, Newman, and Rivard, 1998). The tendons of longissimus thoracis and iliocostalis lumborum pass under the LDF to their sacral and iliac attachments. It appears that the LDF may provide a form of retinacular strapping for the low back musculature. Finally, the abdominal wall and the latissimus dorsi forces add tension to the fascia and stiffness to the spine to prevent specific types of unstable behavior and tissue damage (explained in chapter 5 on spine stability).

Figure 3.46 Stress lines in the LDF indicate that the latissimus dorsi is the dominant force activator—at least in this example.



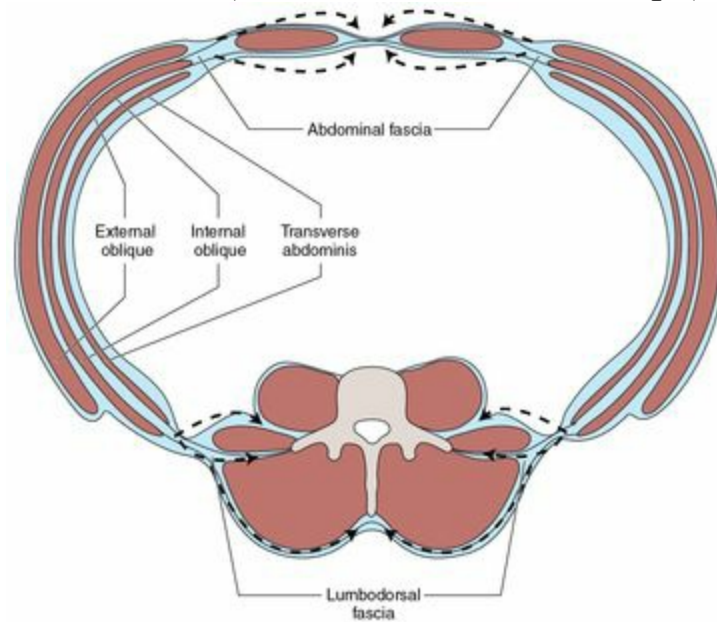
Photo from Stuart McGill

Lumbodorsal Fascia Anatomy: Nature's Back Belt

Clinical Relevance

No evidence justifies specific lifting techniques to involve the LDF for extension of the spine. However, activation of the latissimus dorsi and the deep abdominal obliques contributes stiffening (and stabilizing) forces to the lumbar spine via the fascia (guidelines for activating these muscles are provided in chapter 10). Furthermore, the LDF appears to act as a retinaculum and probably fulfills a proprioceptive function. It is part of a hoop around the abdomen, which consists of the LDF posteriorly, the abdominal fascia anteriorly, and the active abdominal muscles laterally; the three together complete the stabilizing corset (see [figure 3.47](#)). As noted earlier, this also appears to be an important elastic energy storage-recovery device for ballistic athletes, tuned by the obliques.

Figure 3.47 The abdominal fascia, anteriorly, and the LDF, posteriorly, are passive parts of the abdominal hoop. The lateral active musculature (primarily the larger internal oblique and external oblique together with the smaller transverse abdominis) serves to tension the hoop (dashed arrows).

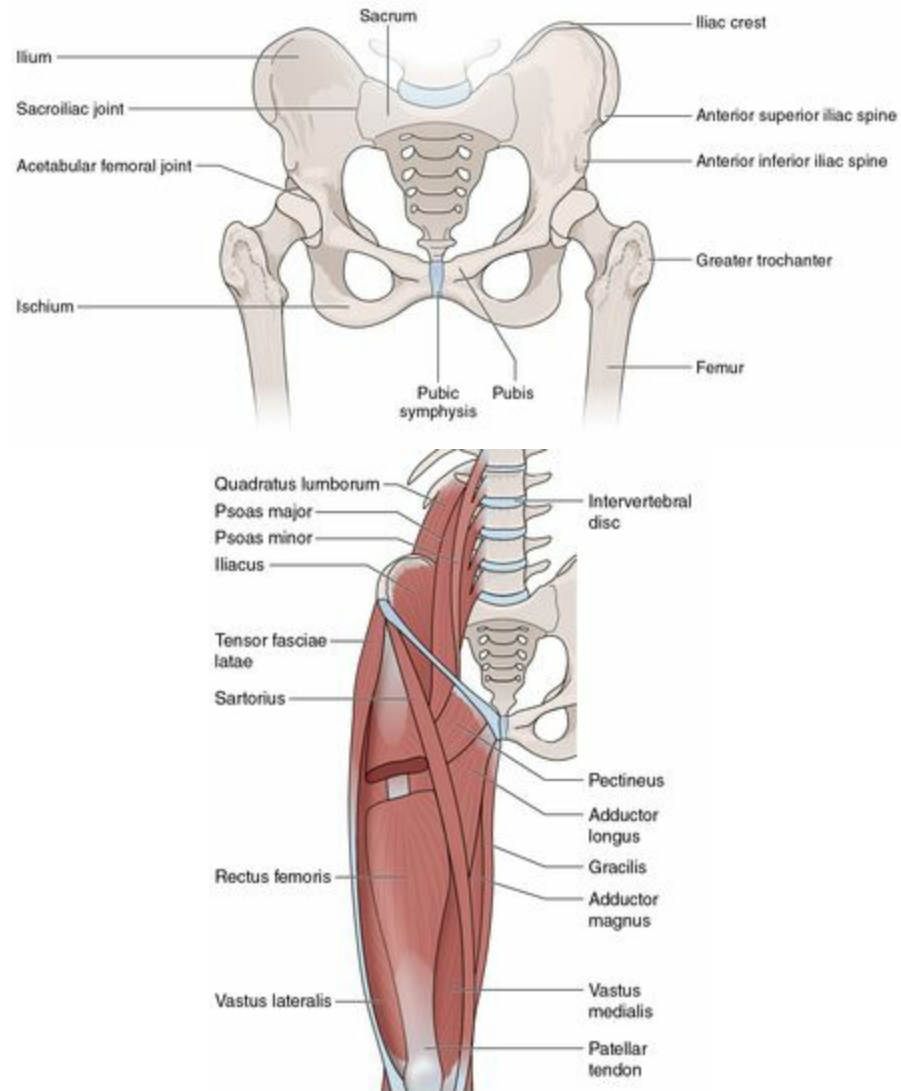


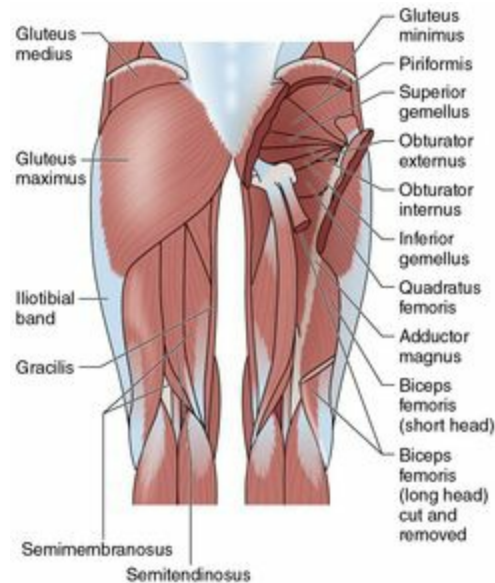
Quick Review of the Pelvis, Hips, and Related Musculature

A healthy back depends on proper function in the pelvis and hips for several reasons. Power is usually generated at the hips, for both performance and safety reasons. Further, the pelvis acts as the platform for the spine. Given the exercises that ensure optimal hip function with spine integration, this brief overview of the relevant anatomy will give context to the exercises later in the book.

The pelvis is usually sectioned into three regions: the ilium, ischium, and pubis. Articulations and small motions occur at the pubis anteriorly and at the sacroiliac joints (see [figure 3.48](#)). Several nerve block studies have shown these to be potential causes of pain. Probably the most relevant factor for exercise-related issues is the architecture of the acetabulum and of the muscles. The hip socket depth and shape influence the ability to squat deeply, which in turn influences the bending moment on the lumbar discs, in particular L5/S1. Interestingly, the shape of the acetabulum tends to be associated with different regions of the world (e.g., Aly and Fuji, 2013). For example, Poland has the highest rate of hip dysplasia (indicative of a shallow acetabulum) (Loder and Skopelja, 2011). But this type of hip shape is best suited for deep squatting—it is no wonder the majority of Olympic weightlifters come from this region of the world. They do not come from Scotland! The reason is that the western European Celtic hip is characterized by a very deep acetabulum (Inoue et al., 2000), which accounts for the very high incidence of anterior femoral hip impingement in that region (Gliedt and Scali, 2012). From a performance perspective the power production in hip extension is very poor out of the bottom of the squat. But the standing stability of the hip is supreme. Tests to define the hip architecture and subsequent impact on therapeutic decisions are discussed later. Training implications are well discussed in the book *Ultimate Back Fitness and Performance* (McGill, 2014).

Figure 3.48 View of the pelvis and hips. Together they function to stabilize and create hip power.





The psoas and iliacus have already been described for their role in hip flexion and stabilization. Gluteus maximus is primarily a hip extensor and external rotator, whereas gluteus medius and gluteus minimus are primarily abductors and external rotators, thus tremendously important for any activity that requires a single-leg stance or a gait with directional change. They assist the spine musculature (such as QL and the obliques) to hold the pelvis up during single-leg support and thus are key players in spine stability during gait. They also externally rotate the femur, which is a functional feature we use to full advantage in the design of squat exercises later in this book.

The gluteus maximus also has fascia connections across the sacroiliac joints that transfer load (Barker et al., 2014). In this way they add stiffness and stability to the pelvic ring and transfer force across the spine–pelvis junction. This architecture also is consistent with the EMG patterns for walking and carrying in which the gluteus on one side is contracted with the quadratus, extensors, and obliques on the other. This is one of the foundations for beginning training progressions with the bird-dog exercise—it challenges the muscles groups with this same synergy. It also surprises many people to learn that the gluteal muscles are efficient extensors of the knee. Even though they do not cross the knee, they work through the skeletal linkage. Here, during a squat, the gluteals extend the hip as it rotates the femur, in turn extending the knee. This unloads the patellar tendon and assists in knee pain in complex hip/knee/back pain syndromes, a technique introduced later. Other “gluteal” muscles known as the deep six (piriformis, obturator internus

and externus, gemellus superior and inferior, and quadratus femoris) together assist in controlling internal and external rotation.

The muscles in the hamstring group (biceps femoris, semitendinosus, semimembranosus) extend the thigh, flex the knee, and perform stabilizing roles over each of these joints. In many upright situations, particularly walking and running, their most important role is in braking with eccentric contraction. Yet in stooped postures, particularly lifting, the hamstring group is very important for their contributions to hip extension. The medial thigh muscles adduct the thigh. The quadriceps extend the knee and provide a patellar tendon tracking function. One member of the “quads,” rectus femoris, crosses the hip joint anteriorly to create hip flexion. Tight hamstrings are often blamed for back troubles, but as is noted several times in this book, this blame is often misdirected. The bulk of evidence supports neither a link between shorter hamstrings and back troubles nor the idea that stretching enhances strength output (e.g., Avela et al., 2003; Fowles, Sale, and MacDougall, 2000) and offers no protective value against injury risk (e.g., Black and Stevens, 2001). Furthermore, what is often attributed to hamstring tightness is actually neural tension so that stretching only worsens the back and radiating leg symptoms. Evidence suggests that hamstring trauma can lead to neural tension (Butler, 1989), although more often the source of the neural tension is associated with a lumbar nerve root or central stenosis.

Training the Hip Musculature

No single exercise can train all hip muscles, although there are some particularly good ones. For example, the one-legged squat is particularly functional, because the quadriceps are active for knee extension and patella control; the hamstrings, for knee stability and hip extension; the gluteals, for hip extension and hip abduction; and the adductors and gluteal deep six, for hip stabilization. Another clinical issue is the peril in training hip flexion power given the associated loading of the lumbar spine. This is usually reserved for those who no longer have pain, because training hip flexion often retards progress toward the elimination of pain. It is usually wise to build extensive spine stability prior to progressing to building hip power.

Hip mobility is essential for sparing the back from bending movement that causes stress, together with assisting with power transfer to the legs. The question is how best to restore and maintain hip mobility. Moreside and McGill (2011, 2012) performed a suite of studies showing that stretching targeting muscle, nerve, joint capsule, and fascia can indeed enhance the range of motion. However, they found that enhancing core stiffness enhanced hip mobility. Further, simply adding to the hip joint range of motion did not translate into more hip motion during a variety of functional activities. This is because the movement engrams, or encoded patterns of motion, need to be changed. The implication is that if more movement in the hips is desired, a hip mobility program needs follow-up exercise designed to rewrite the movement engrams to incorporate the newfound movement.

Considering these muscles reveals the tremendous number of them that cross the hip joint. Collectively, they can create significant power, and they can direct force in a powerful manner in many directions. In many cases they are the major power source for functional performance. Specifically, they create and direct force while changing length at rapid speed. Many of them cross both the hip and knee joint, indicating that functional training must involve tasks that challenge both joints at different velocities. Further, this training must incorporate lateral motion and rotational motion in the transverse plane. All function involving power generation across the hip relies on the ability of the core to stiffen and stabilize. The stiffer the core, the greater the power transfer to the limbs.

Clinically Relevant Aspects of Pain and Anatomic Structure

Recall from the introduction to this book that tissue damage can alter the biomechanics of a spinal joint, and that once the biomechanics have changed, any innervated tissue can be the candidate for symptoms. Patients do not get used to pain, nor is it in their heads, as poorly trained doctors often claim. Pain biology clearly describes pathways beginning with mechanical factors such as disc bulges causing nerve root compression. Pain originates with the free nerve endings of the various pain receptors that typically form small nerve fibers. As Guyton (1981) noted, not all of the small fibers originate in pain receptors: some originate in organs sensitive to temperature, pressure, or other sensations. Pain also may be initiated at higher levels in the pain pathway: Howe, Loeser, and Calvin (1977) demonstrated that mechanical pressures on the dorsal root ganglion produce discharges for up to 25 minutes following the removal of the mechanical pressure. In addition, Cavanaugh (1995) showed that nerve endings are sensitive to chemical mediators released during tissue damage and inflammation. Some studies have attempted to examine inflammatory processes using the injection of chemicals. For example, Ozaktay and colleagues (1994) injected carrageenan into the region of nerve receptors around the facet joints of rabbits and reported that the discharge from the pressure-sensitive neurons lasted 3 hours. This finding suggests that tissue damage producing inflammatory processes may contribute to a long-lasting muscle spasm.

Direct mechanical contact and pressure on the nerve roots located at each level of the spine create changes that cause pain and pain sensitivity. Hubbard, Chen, and Winkelstein (2007) measured the allodynia (increase in pain sensitivity) from mechanical pressure applied to nerve roots simulating a disc bulge in an animal model (unaffected by compensation, an employer, an emotional state, and so on.). They investigated the increased neuropeptides that cause this disabling and disruptive pain and their link to load magnitude and duration. Others have documented mechanical compression causing edema inside the nerve root sheaths (Olmarker, Rydevik, and Holm, 1989; Winkelstein et al., 2001), which results in bilateral sensitivity (Hubbard and Winkelstein, 2005) and central sensitization. This is similar to hitting your thumb with a hammer. Your tissues are damaged and sensitized. Hit it again

tomorrow and it will be more painful. In two days, hitting your thumb will be excruciating and disabling. This is what many back pain sufferers experience because of the way they load and move their spines, unknowingly hitting their spines with pain hammers.

Bogduk (1983) provided an excellent review of the innervation of lumbar tissues. For example, the facet is well innervated with a variety of low- and high-threshold nerves, suggesting both pain and nociceptive functions. Free nerve endings also have been observed around the superficial layers of intervertebral discs. With injury, nerves sprout into the damaged region. In summary, mechanical factors cause disabling pain that lasts. Addressing this pain by changing the mechanical cause is the only way to have success with many back pain cases.

Tissue-Specific Types of Pain

I have had some personal experience with direct mechanical irritation of specific low back tissues and the resultant pain. Admittedly, these results are limited, given the single subject (myself) and the subjective nature of the observations, but I believe they are worth reporting nonetheless. I obtained these insights from indwelling EMG experiments in which needles were used to implant fine wire EMG electrodes in the psoas, QL, multifidus, and three layers of the abdominal wall. Many people have experienced the burning sensation as the needle penetrates the skin. This is cutaneous pain, because the application of ice and a hot material feels similar. As the needle applies pressure to, and punctures through, the LDF, the pain is felt as a scraping sensation and sometimes as an electric current. The same sort of pain is felt as the needle progresses through the various sheaths between the layers of muscle of the abdominal wall. It is interesting to note that people with fibromyalgia sometimes report this scratchy type of muscularly located pain—it is very consistent with epimysium and fascia irritation. Once the needle was inside the muscle, no pain was perceived, just an occasional feeling of mechanical pressure. As the needle touched the peritoneum of the abdominal cavity in any location, a general intestinally sick feeling was produced in the abdominal region, focused anteriorly to a small area just below the naval. As the needle touched the bone of the vertebra, even with very light pressure, a very pointed and “boring” pain occurred, similar to the pain experienced on

being kicked in the shins. Once again, these are the experiences of a single person. Nonetheless, they do provide crude qualitative insight into the types of pain produced in specific tissues.

Can Pain Descriptors Provide a Reliable Diagnosis?

Pain is clearly produced from tissue irritation, particularly mechanical overload. Some have argued that some tissues may or may not be candidates for sources of pain based on the presence or absence of nociceptive nerve endings. This may be a diversionary argument. If the overload is sufficient to damage tissue and produce biomechanical change in the joint, then the loading patterns of other tissues are disturbed. Thus, even though one tissue may not be capable of producing pain, if it is damaged sufficiently to shift load to another suitably innervated tissue, pain may result. For example, innervated annulus and disc end plates may be sources of pain as a consequence of end-plate fracture or annular herniation. But end-plate fracture can cause significant disc height loss, which can lead to nerve entrapment, complex joint instability, subsequent facet overload, and so on. Evidence from our lab (Freeman et al., 2013) showed that induced hip pain inhibited gluteal activation, which changes the loading on the spine and anterior hip joint capsule and labrum. Once the biomechanics of the joint have been altered, it is no longer fruitful to attempt to diagnose specific tissue damage; the picture is complex. Functional diagnosis is the only feasible option.

A Final Note

The discussion in this chapter assumes a rudimentary anatomical and biomechanical knowledge. Using this foundation, some anatomical features were reviewed that are not often considered or discussed in classic anatomical texts. A serious discussion of anatomy must involve function and, by extension, must consider biomechanics and motor control. Hopefully, the functional discussions throughout this chapter have stimulated you to give more consideration to the architecture of the lumbar spine. The challenge for the scientist and clinician alike is to become conversant with the functional implications of the anatomy, which will guide decisions to develop the most appropriate prevention programs for the uninjured and the best treatments for patients.